

1. **Course Code:** CSE-2340
2. **Course Title:** Software Development 1
3. **Credit Hours:** 2
4. **Contact Hours:** 4 lab hours per week
5. **Type:** Core, Engineering
6. **Prerequisite:** CSE-1221 / CSE-1222
7. **Co-requisite:** -
8. **Instructor's and Class Schedule and Locations**

Instructor: Md. Mahadi Hassan
Associate professor, Dept. of CSE.

9. General Overview:

This course gives students experience designing, implementing, testing, and debugging large programs. Students will also get advanced python programming experience; covering topics such as inheritance, multithreading, networking, database programming, and web development.

10. Course Objectives:

- To develop knowledge of phases in software development.
- To develop good quality software and able to maintain quality of software.
- To know the team required for project management.
- To develop knowledge of tools available for software development.
- Knowledge of testing and maintain robustness of software

11. Course Outcomes (COs):

Upon successful completion of this course, students will be able to:

#	CO Description	Weightage (%)
CO1	To understand about designing model and practical implementation.	30
CO2	To take decision of project planning on the basis of cost evaluation.	20
CO3	To understand risk identification and management.	10
CO4	To use various tools for software design development.	25
CO5	To understand importance of quality of software.	15

12. Mapping of CO-PO:

#	CO Description	POs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CO 1	To understand about designing model and practical implementation.	PO1	Cognitive/Understand	Lecture, Class discussion, Assignment, Lab work	Assignment, Exam, Lab performance, Project
CO 2	To take decision of project planning on the basis of cost evaluation.	PO2	Cognitive/Analyze	Lecture, Class discussion, Assignment, Lab work, Note	Assignment, Exam, Lab performance, Project

#	CO Description	POs	Bloom's Taxonomy Domain/Level	Delivery Methods and activities	Assessment Tools
CO 3	To understand risk identification and management.	PO1	Cognitive/Understand	Lecture, Class discussion, Assignment, Lab work	Assignment, Exam, Lab performance, Project
CO 4	To use various tools for software design development.	PO3 PO5	Cognitive/Apply	Lecture, Class discussion, Assignment, Lab work	Assignment, Exam, Lab performance, Project
CO 5	To understand importance of quality of software.	PO1	Cognitive/Understand	Lecture, Class discussion, Assignment, Lab work	Assignment, Exam, Lab performance, Project

13. Resources:

Text Book

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1	Brian Allbee	Hands-On Software Engineering with Python	-	amazon		978-1788622011
2	Jason Grembi	Secure Software Development: A Security Programmer's Guide	1st Edition			ISBN-13: 978-1418065478 ISBN-10: 1418065471
3	Ramez Elmasri, Shamkant B. Navathe	Fundamentals of database systems	6 th	Pearson Education	2011	10: 0-136-08620-9

Reference Books:

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
1.	J. Date, A. Kannan and S. Swamynathan	An Introduction to Database Systems	8 th	Pearson Education	2009	
2.	Russ Miles, Dan Pilone	Head First Software Development	3rd Edition	O'Reilly Media, Inc.	2017	9780596527358

#	Name of Authors	Title of Book	Edition	Publisher's Name	Year	ISBN
3.	Hans van Vliet	Software Engineering: Principles and Practice	2 nd Edition	Wiley	2007	9780471975083

Online Resources:

- I) <https://www.geeksforgeeks.org/c-programming-language/?ref=leftbar>
- II) <https://www.programiz.com/>
- III) <https://www.tutorialspoint.com/>
- IV) <https://w3schools.com/>

14. Course Assessment Pattern:

Bloom's taxonomy domain/level	Evaluation out of 100 marks			
	CIE (50 marks)		SEE (50 marks)	Total (100)
	Attendance (10)	Continuous Assessment (40)	Final Exam (50)	Total (100)
Cognitive/Remember	-	-	-	
Cognitive/Understand	-	20	20	40
Cognitive/Apply	-	10	15	25
Cognitive/Analyze	-	10	15	25
Cognitive/Evaluation	-	-	-	-
Cognitive/Create	-	-	-	-
Affective/Responding	10	-	-	10

15. Grading Policy:

As per IIUC grading policy

16. Weekly Activity Plan:

Week	Activities	Topics
1	Lecture (4)	Introduction of the course, Software Processes , Software Project Management, Software development life cycle
2	Lecture (4)	Gathering requirements: knowing what the customer wants, Project planning: planning for success, User stories and tasks: getting to the real work.
3	Lecture (4) Class test 1	OOP basic introduction, Control statements, Understanding encapsulations
4	Lecture (4)	Data type in python, data type conversion, Date, Time, Method, Parameterized method
5	Lecture (4)	Polymorphism (Method overloading) Constructor, Difference between Constructor And Method
6	Lecture (4) Project Proposal	Constructor overloading, constructor chaining. Data structure in python (Array, Array List) and their use.

7	Lecture (4)	Inheritance basics, Method overriding, Super class, Sub class, Abstract class & methods, Interface basics
8	Lecture (4)	Interface vs. multiple inheritance Difference b/w interface & Abstract class, Framework introduction (installations & procedures)
9	Lecture (4)	Exceptions & Exceptions handling, Software design, Take user input
10	Lecture (4) Class test 2	Take user input Save user information and view into a table. MSSQL introduction, SQL server installation, Database table create.
11	Lecture (4)	Data insert, delete, update (without SQL query & with query) SQL connections create with python project.
12	Lecture (4)	Data handling from <i>User Interface</i> , Specifications, Criteria for specification
13	Lecture (4)	Requirement analysis, Requirement specifications
14	Lecture (4)	Modelling & architecture, Software Implementation
15	Lecture (4) Project Showcase	Software maintenance, Documentation techniques.